

BHARATI MUDIGOUDRA *Aspiring AI Engineer*

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🌐 LinkedIn 🐙 GitHub 🔗 Portfolio

👤 CAREER OBJECTIVE

AI Engineer with hands-on experience building applications powered by Large Language Models (LLMs), Retrieval Augmented Generation (RAG) pipelines, agentic workflows, and computer vision systems through internships and independent projects. Skilled in Python, FastAPI, LangChain, LangGraph, and vector databases, with projects deployed on AWS (Bedrock, Lambda, API Gateway) and containerized with Docker. Also brings a year of professional experience supporting US insurance clients - a strong fit for AI roles in insurance technology

🧠 SKILLS

Generative AI — LLMs, RAG, Prompt Engineering, Embeddings, Vector Search, AI Agents, Agentic Workflows, NL2SQL, LoRA, VLM, ReAct, MCP

AI/ML Engineering — Machine Learning, Feature Engineering, Model Selection, Classification, Regression, Clustering, Ensemble Learning, Model Evaluation

Computer Vision & NLP — Object Detection, Image Classification, Segmentation, YOLO, Transformers, LSTM, RNN

AI Frameworks & Development — Python, FastAPI, Flask, LangChain, LangGraph, LlamaIndex, PyTorch, TensorFlow, Scikit-learn, Pandas, NumPy

Data & Vector Infrastructure — SQL, MySQL, MongoDB, Pinecone, Weaviate, ChromaDB, FAISS

Cloud & MLOps — AWS, GCP, Docker, GitHub Actions, CI/CD, Kubernetes

AI Evaluation & Monitoring — LangSmith, Ragas, DeepEval

📁 WORK EXPERIENCE

Associate, SDU 06/2025 – 05/2026 | Bangalore
ResourcePro

- Analyzed and validated business data while ensuring high accuracy and quality.
- Followed SOPs and established workflows to process tasks efficiently.
- Investigated data discrepancies and supported issue resolution through detailed analysis.
- Prepared reports and collaborated with stakeholders to support business operations.
- Contributed to process improvements that enhanced operational efficiency and productivity.

Data Analyst Intern 10/2024 – 01/2025 | Bangalore
AI4SEE Pvt Ltd

- Worked on Data collection, augmentation, cleaning and annotated the data using RoboFlow for YOLOV10 model.
- Developed a Real-time Object Detection System integrating YOLOv10 and DERT Models, enhancing detection accuracy using Non-Maximum Suppression (NMS) and Intersection over Union (IoU).
- Researched garbage classification (biodegradable & non-biodegradable) to advance waste management automation.
- Worked on Graph Neural Networks (GNNs) for traffic prediction, improving real-time traffic forecasting accuracy

Data Science Intern 06/2024 – 10/2024 | Bangalore
iNeuron Intelligence Pvt Ltd

- Developed an AI-powered data analysis system using LLMs via Groq API, automating insights generation.
- Implemented real-time processing & predictive modeling, ensuring quick adaptation to changing data patterns.
- Designed scalable and secure data pipelines, enhancing business intelligence applications.
- Explored various LLM's performance for the data analysis.

EDUCATION

MCA

Bangalore University

MCA was completed with distinction, achieving an overall CGPA of **8.05**

02/2023 – 11/2024 | Nelamangala, India

BCA

Kuvempu University

BCA was completed with First Class, achieving an overall GPA of **74.5%**

06/2019 – 11/2022 | Shivamogga, India

PROJECTS

ZeroQueue - AI Gift-Card Support Copilot

- Built an AI gift-card support copilot during the HiDevs YC Fall 2026 "Moss Zero-Latency Builder Sprint" hackathon; live at <https://zeroqueue-hubble.onrender.com/> .
- Developed a Python and FastAPI backend with a RAG pipeline over a 100-brand support knowledge base indexed in Moss (minilm embeddings, in-memory hybrid search, ~10 ms retrieval), returning cited answers with confidence scoring.
- Implemented voucher image upload with Tesseract OCR (installed via Dockerfile) and automatic clarifying questions for incomplete requests.
- Integrated human handoff through the Intercom REST API, including contact and conversation creation, team assignment, and webhook-driven human replies shown live in the chat.
- Containerized with Docker and deployed on Render, with 42 pytest tests covering the backend.

Tech Stack: Python, FastAPI, RAG, Tesseract OCR, Intercom REST API, Docker, Render, Pytest

Medical Chatbot using Gen AI

- Built a RAG-based medical chatbot that retrieves relevant information from a medical knowledge base before generating responses with LLaMA 3, reducing reliance on unsupported LLM-generated answers.
- Designed the retrieval and generation pipeline using LangChain and Pinecone, using vector search to provide relevant context to the LLM for each user query.
- Developed Flask REST APIs for query processing, connecting the retrieval layer, LLM inference, and chatbot application into a reusable backend service.
- Deployed the application using AWS Bedrock, Lambda, and API Gateway, creating a cloud-based architecture for serving AI-powered queries through APIs.
- Applied prompt engineering and vector-search optimization to improve consistency of generated responses.

Tech Stack: Python, Flask, LangChain, LLaMA 3, Pinecone, AWS Bedrock, Lambda, API Gateway

Fire & Smoke Detection

- Prepared and configured the custom YOLO dataset with annotated fire and smoke instances and train/validation/test data splits.
- Fine-tuned YOLO11 on the custom dataset and generated trained weights for image and video inference.
- Achieved 92.2% mAP@0.5, with Fire mAP@0.5 = 92.6% and Smoke mAP@0.5 = 91.8%.
- Evaluated model performance using confusion matrices and confidence curves, with an overall F1-score of 0.89 at 0.441 confidence and high precision at optimized confidence thresholds.
- Developed OpenCV-based image/video inference, detecting fire and smoke frame-by-frame, drawing bounding boxes with confidence scores, and generating annotated outputs for continuous monitoring.

CERTIFICATE

Generative AI Mastery & Full stack Data Science — Euron

Fundamentals of Generative AI — Microsoft

Soft Skills & Personality Development — CRACK-ED

Artificial Intelligence — IBM